

Instrument & Tooling Calibration

Standard Operating Procedure

Doc ID	SOP-QA-003
Version	2.0
Effective Date	2026-02-01
Owner	Maintenance Department
Scope	Torque tools, dial indicators, tool offset registers, and inspection gauges across all lines

1. Purpose

Defines the calibration frequency and method for measurement and torque tooling to prevent DT-06 calibration-drift downtime and ensure part quality conformance.

2. Required PPE

- Safety glasses

3. Tools & Materials

- Calibrated reference standards (traceable)
- Torque tester
- Calibration log sheet / CMMS terminal

4. Procedure

- 1 Confirm the calibration due date on the asset tag before use each shift.
- 2 For torque tools: test at 20%, 60%, and 100% of rated scale against the torque tester; record readings.
- 3 For dial indicators/gauges: zero against a certified reference standard and check repeatability across 3 readings.
- 4 If deviation exceeds $\pm 3\%$ of nominal, remove the tool from service and tag 'Out of Calibration'.
- 5 Send out-of-tolerance tools for recalibration or replacement; do not attempt field adjustment of torque clutches.
- 6 Update the calibration record with new due date, technician ID, and result.

5. Associated Downtime Code

Code	Description	Severity	Typical Duration
DT-06	Calibration Drift	Moderate	20 min

■ Never use a tool flagged 'Out of Calibration' for production — doing so risks undetected quality escapes.